No. of Experiment - 01

Name of the experiment: Determination of Young's modulus by the flexure of a beam (bending method).

Theory:

Young's modulus, also known as the elastic modulus, depicts the stiffness of a material. It is the ratio of stress to strain, where **stress** is the applied force per unit area, and **strain** is a ratio of the material's change in length due to the stress vs. the initial length. **Hard and brittle materials** like ceramics have a very high young's modulus because they can sustain a lot of stress before "yielding" (in the case of brittle materials, breaking). **Metals** are more flexible, so they begin to yield at a lower stress. **Plastics** tend to have very low modulus because they are often rather flexible. Even brittle plastics have low moduli compared to metals because they are weaker overall.

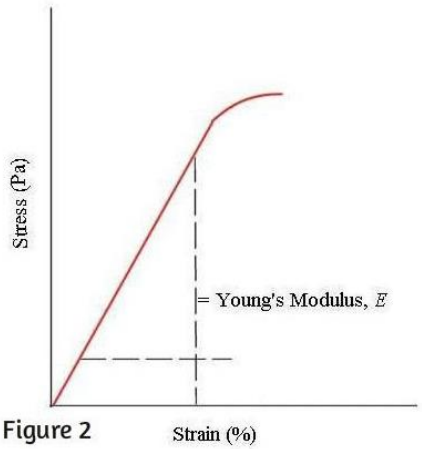


Figure 2

For 2 pieces of same shape and size but different materials loaded under same force, the one with **higher Young's modulus** will deform less than the other. If these are elongated to same extent then, the one with higher Young's modulus will be subjected to **higher load**. We are assuming that there is no permanent deformation in either of the two pieces.

- The Young's modulus directly Measures the **stiffness** of the Solid material.
- It defines the **relationship** between stress (force per unit area) and strain (proportional deformation) in a material.
- Young's modulus (E) is a measure of the ability of a material to **withstand changes** in length when under length wise tension or compression.
- It predicts how much a **material sample** extends under tension or shortens under compression.

Young's modulus or Modulus of Elasticity which is a *material Constant* Value

If a light bar of breadth b and depth d is placed horizontally on two Knife-edges separated by a distance l, and a load of mass M, applied at the mid-point of the bar, produces a

Depression y_0 of the bar, then Young's modulus Y of the material of the bar is given by

$$Y = \frac{Mgl^3}{4bd^3y_0} \qquad \text{---------- (i)}$$

Where, g is the acceleration due to gravity. This is the working formula of the experiment, and is valid so long as the slope of the bar at any point with respect to the unstrained position is much less than unity. Here Y is determined by measuring the quantities b, d, $L$and the mean depression l corresponding to a load m. If b, d, l and y_0 are measured in cm, m in gm, g is expressed in cm/sec^2, and then Y is obtained in dyne/cm^2.

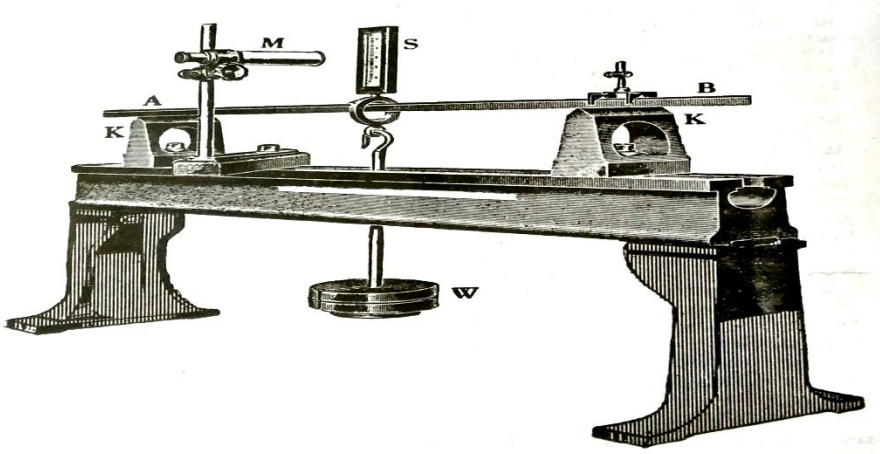


Fig.2.14

Required Apparatus:

1. Rectangular beam
2. Suitable weights
3. Vernier calipers
4. Screw gauge
5. Meter scale and
6. Spherometer, etc.

Procedure:

1. Measure the length of bar between knife edges using meter scale. This will give you value of l.
2. Find out the least count of screw gauge and zero error in it.
3. Using screw gauge, measure the thickness of bar/beam. It will provide the value of d.
4. Find out the least count and zero error of vernier calipers.
5. Using vernier calipers, measure the breadth of beam. It will provide the value of b.
6. Find the least count of spherometer.
7. Suspend the hanger with a graduated scale attached to it, on the mid-point of the beam. Now rotate the spherometer. As soon as it touches the beam, the bulb glows. Stop the rotation of spherometer and note its reading. This gives zero mass depression.
8. Now load 500 gm on hanger. At this position, the bulb will not glow. Further rotate the spherometer till the bulb just glows. Stop the rotation of spherometer and note its reading. This gives depression for 500gm.
9. Increase the mass on hanger till 2500gm in interval of 500gm and note the spherometer readings at the situation of bulb glow. This will give you the depressions at different masses.

Observation Table:

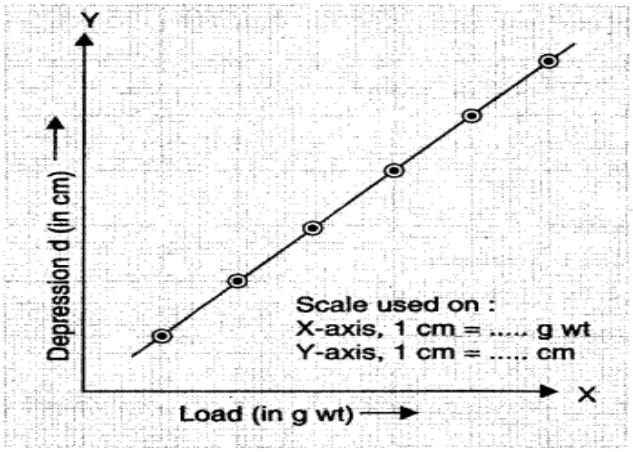
1. Length of beam between knife edges (l)=…..... cm
2. Least count of screw gauge = $\frac{pitch}{Number\ of\ \text{divisions on circular scale}}$ =…….cm

Table-1 for depression table

No. of observations	Load, M in gm	Readings for load increasing				Readings for load decreasing				Mean reading in cm	Mean depression y_0
		Main scale reading	Vernier scale divn. (V.D)	Vernier scale reading (V.D×V.C)	Total (cm)	Main scale reading	Vernier scale divn. (V.D)	Vernier scale reading (V.D×V.C)	Total (cm)		

Calculation:

We will get a straight line from the graph.....



Graph between load and depression.

Length, l = ...cm

Breadth, b = M.S.R + V.S.R × V.C = ...cm

Thickness, d = L.S.R + C.S.R×L.C - (±e)×L.C = ...cm

$$\therefore\ Y = \frac{Mgl^3}{4bd^3y_0} = \text{.....dyne/cm}^2$$

Result: The Young's modulus of the beam is, Y =..... dyne/cm^2. Standard value is $(10-13)\times 10^{11}$ dyne/cm^2.

Percent Error: $\frac{Standard\ value \sim Calculated\ value}{Standard\ value} \times 100 = \cdots \%$

Precautions:

(i) In the working formula, Young"s modulus Y is the function of the cube of length (L) and depth (d), they should be measured very carefully, otherwise a large error will occur in the value of Y.

(ii) The beam should be made horizontal and the loading should be made exactly at the middle of the bar.

(iii) As the breaking load of the bar is very high, the maximum load on the hanger is usually far below the breaking load. Hence it is unnecessary to find the breaking load of the bar.

(iv) To minimize the error in measuring Y, it is available to find Y for three different lengths of the bar and the length of the bar should not be below 80 cm.

(v) While measuring depression, to avoid back-lash error the microscope screw should allows be rotated in the same direction.

(vi) After adding a load to the hanger, reading is to be taken after waiting for sometime show that the depression is complete.

No. of Experiment - 02

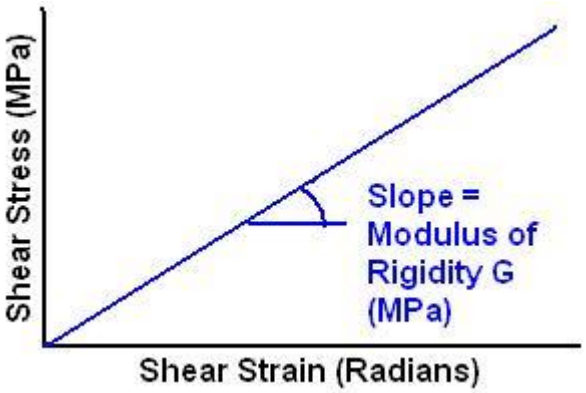
Name of the Experiment: Determination of the **modulus of rigidity** of a wire by statical method.

Theory: The rigidity modulus or shear modulus is a measure of the ability of a material to **resist transverse deformation** and is a valid index of elastic behavior only for small deformations, after which the material is able to return to its original configuration.

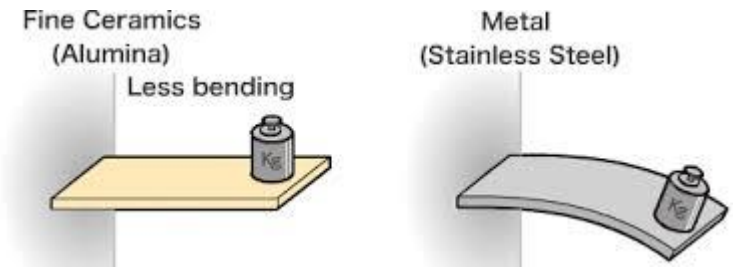
Within elastic limit, **Young's modulus**, $Y = \frac{Longitudianal\ stress}{Longitudinal\ strain}$

Modulus of rigidity: It is defined as the ratio of shear stress to the corresponding shear strain within elastic limit. i.e.

Rigidity modulus, $\eta = \frac{Shear\ stress}{Shear\ strain}$



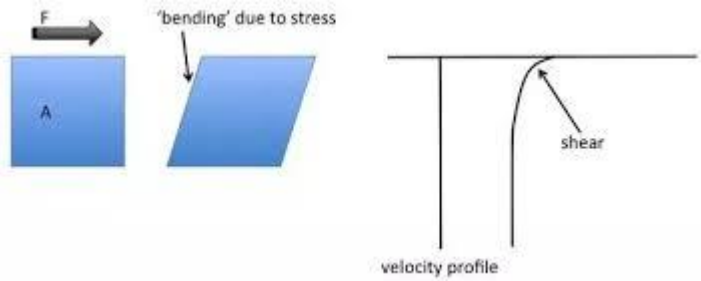
Rigidity, also known as "stiffness," is generally measured using Young's modulus. It can be defined as the "force necessary to bend a material to a given degree.

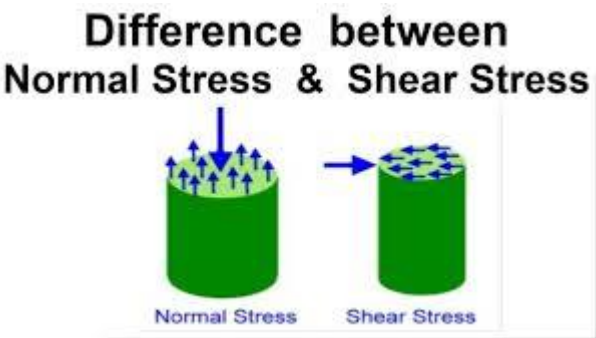


$\eta_{ceramics} > \eta_{metals}$

Fine Ceramics are highly rigid materials, according to Young's modulus. This makes their machining accuracy high enough to enable them to be used for high-precision parts.

Shear stress:





Normal stress: The force acting perpendicular to the area of contact is normal stress = $\frac{F}{A}$

Where, F = Applied force

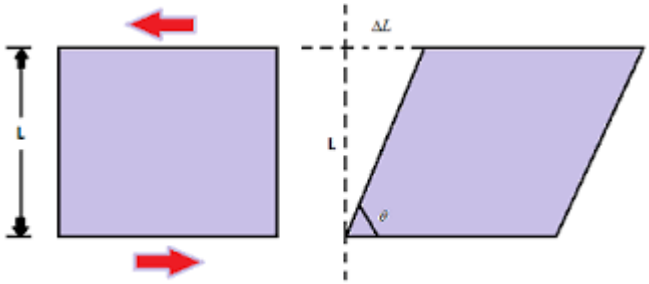
A = Cross-sectional area

Example: Cutting a cake with a knife. In physics, the cake experiences normal stress.

Shear stress: The force acting parallel to the area is shear stress.

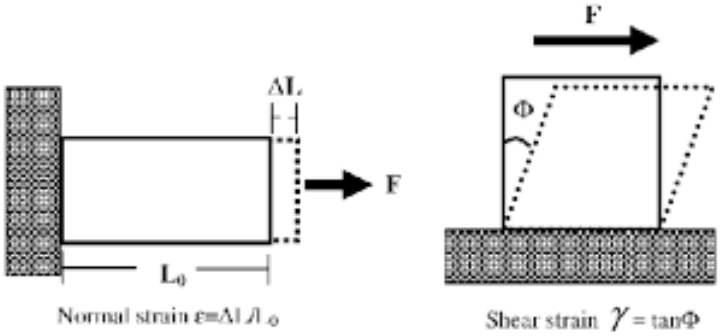
Example: Cutting a cake through cross-sectional area.
Again, Breaking a wooden stick by leaning it against a wall and stepping on it really hand and break it. What made it broken is shear stress. It is occurred in different loading conditions.

Shear strain:



Shear Strain

Strain = $\frac{\Delta L}{L}$ (Fixed base)
Where, ΔL = Change in length
L = Original length
Applied force is parallel to the area.



Shear strain is defined as the ratio of deformation over the nominal distance between the deformed and un-deformed planes.

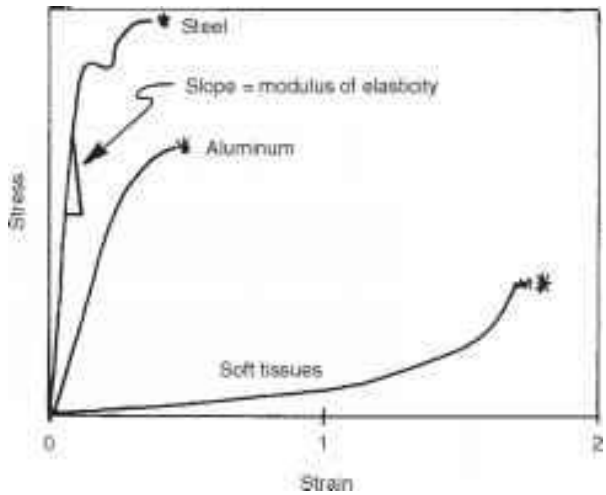
$$\text{Shear strain} = \tan\theta$$
$$= \frac{Opposite}{Ajacent}$$
$$= \frac{\Delta L}{L}$$

The following formula is used for the determination of the modulus of rigidity of a given experimental rod.

$$\eta = \frac{360 \times mgd(l_1 - l_2)}{\pi^2 r^4 (\theta_1 - \theta_2)}$$

Where, m = load suspended on each cord,
d = diameter of the heavy cylinder/flywheel
$l_1 - l_2$ = distance between two circular scales
$\theta_1 - \theta_2$= difference between deflections measured in circular scales
r = radius of the experimental rod
g = acceleration due to gravity

$\frac{\theta_1}{l_1} = \frac{\theta_2}{l_2}$. i.e. Greater the distance from the clamped end, the greater will be the twist produced. Where, θ_1 & θ_2 are the corresponding twists produced for the given equal load.



Required Apparatus:

1. Modulus of rigidity apparatus (Barton's apparatus)
2. Slide Calipers
3. Screw Gauge
4. Weights
5. Meter Scale, etc.

Procedure:

1. Fix the both scales and pointers at different places such that pointers should indicate 0-0 on scales in absence of load (Note: at least pointer on one side must indicate zero). Now measure distance between scales.
2. Place equal masses (start from 0gm in interval of 500gm) on each pan and read deflection on both scale and on both side. The scale which is nearer to cylinder will provide θ_1 value while other will give θ_2 value.
3. Go on increasing masses on each pan by equal amounts (from 0 to 2.5kg in interval of 0.5kg) and note the corresponding deflections on both scales on both sides.
4. Go on increasing masses on each pan by equal amounts (from 0 to 2.5kg in interval of 0.5kg) and note the corresponding deflections on both scales on both sides.
5. Now decreases the masses on pans in the same interval and note the corresponding readings on scales for the case of load decreasing.
6. Take mean of all four readings of θ_1 which is noted for the case of load increasing and decreasing. Similarly, do it also for θ_2.
7. After it, calculate θ_1-θ_2 for each masses. Using these values, find the angle of twist (θ_1-θ_2) for 1.5 kg (it can be obtained by taking difference between 1st & 4th, 2nd & 5th). Find out the

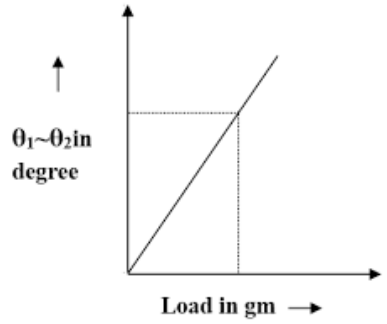
least count of screw gauge and zero error in it. Using screw gauge, measure the diameter of wire. Its half will provide the value of radius (r) of wire/experimental rod.

8. Find out the least count and zero error of Vernier calipers. Using Vernier calipers, measure the diameter of cylinder (D).
9. Put all the values in the formula and calculate it by log method.

Observation Table:

Reading for load-twist graph

No. of Obs.	Load in gm	Pointer readings in degrees				Mean pointer reading		Mean twist in degrees $(\theta_1 \sim \theta_2)$
		Scale S_1		Scale S_2		(θ_1) in degree	(θ_2)in degree	
		Load increasing	Load decreasing	Load increasing	Load decreasing			
1								
2								
3								
4								
5								



Calculation:

(1) Distance between two circular scales $(l_2 - l_1)$ =……..cm

(2) Least count of screw gauge= $\frac{Pitch}{\text{Number of divisions on circular scale}}$ =……..cm

(6) Radius of wire (r) = $d_w/2$=……..cm

(7) Least count of Vernier calipers = $\frac{Value\ of\ one\ division\ on\ main\ scale}{Number\ of\ divisions\ on\ vernier\ scale}$ ……..cm

Put all the values in the above formula and solve

$$\eta = \frac{360 \times mgd(l_1 - l_2)}{\pi^2 r^4((\theta_1 \sim \theta_2)\ )} = \frac{360 \times gd(l_1 - l_2)}{\pi^2 r^4} \times \frac{m}{((\theta_1 \sim \theta_2)\ )}$$

Here, m and $(\theta_1 \sim \theta_2)$ are determined from graph.

Results: The modulus of rigidity of given experimental rod/wire material =……. dyne/cm^2. Standard value is $(7.9 - 8.9) \times 10^{11}$ dyne/cm^2.

Percent Error: $\frac{Standard\ value \sim Calculated\ value}{Standard\ value} \times 100 = \cdots \%$

Precautions:

1. There should be friction in pulleys.
2. The cord wounded on cylinder should be thin and strong.
3. Load should be increased and decreased gently.
4. Load should not exceed the limit of elasticity. 5. To avoid the backless error, the circular scale of screw gauge should be moved in one direction in measurement of diameter of experimental rod/wire.

No. of Experiment- 03

Name of the experiment: Determination of spring constant and effective mass of a loaded spring.

Theory: Hook's law states that the force needed to compress or extend a spring is directly proportional to the distance you stress/compress it but in the opposite direction. It is applicable in spring.

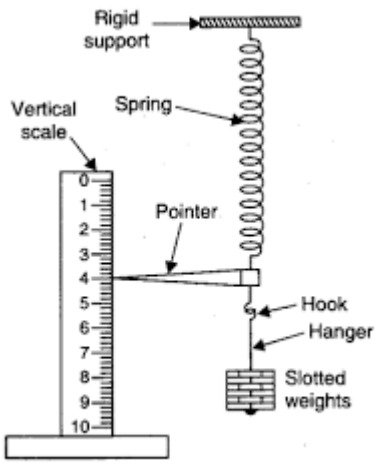


Fig. Extension of spring.

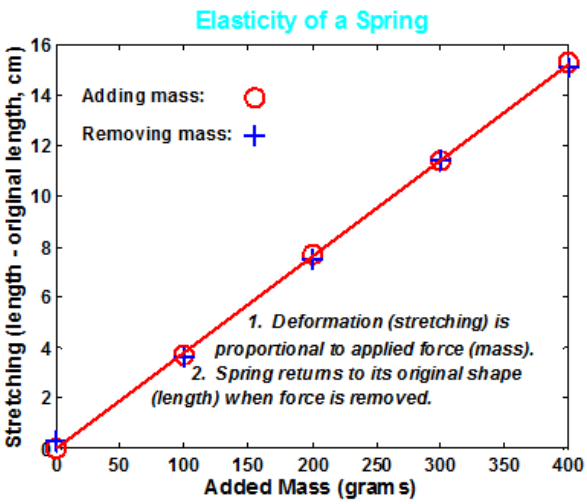
$$F_{ext} = kx$$
$$F_s = -kx$$

Where, F_s is the spring force/restoring force

$$F = kx$$
$$mg = kx$$
$$k = \frac{mg}{x}$$

This is the formula of spring constant.

From m vs x graph



In spring, potential energy (mechanical energy) is stored. It converts into kinetic energy.
All motions of the spring have same period of oscillations.

$$\omega^2 = \frac{k}{m} \text{ or } \omega = \sqrt{\frac{k}{m}}$$

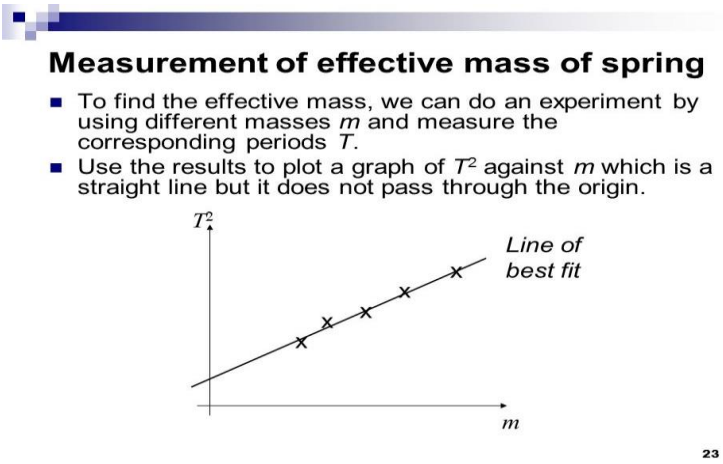
$$T = 2\pi\sqrt{\frac{m}{k}}$$

(i) For large k, ω = large (rapid oscillations) and small T.
(ii) For a large mass m, ω = small (sluggish oscillations) and T is large.
Less spring constant, less oscillations.

Effective mass: In a real spring mass system, the spring has a non negligible mass m. Since not not all of the spring's length moves at the same velocity v as the suspended mass m_0, its kinetic energy is not equal to $\frac{1}{2}mv^2$. As such, m cannot be simply added to m_0 to determine T and the effective mass of the spring is defined as the mass that needs to be added with m_0 to correctly predict the behavior of the system. [i.e. For accurate SHM]

$$T = 2\pi\sqrt{\frac{m+m_0}{k}}$$

Where, m is the effective mass.
$m = \frac{1}{3}M$; where M is the true mass of the spring.
From T^2 vs m graph we get m.



Required Apparatus:

1. A spiral spring,
2. Convenient masses with hanging arrangement,
3. Weighing balance,
4. Stopwatch
5. Clamp or a hook attached to a rigid framework of heavy metal rods etc.

Observation Table:

Table-1 for the determination of spring constant

No. of Obs.	Load (m) in gm	Extension (x) in cm		Mean	Actual value
		Load increasing	Load decreasing		

Table-2 for the determination of effective mass

No. of obs.	Load (m) in gram	Time (t) for 20 oscillations	Period (T) $T = \frac{t}{20}$ in sec	T^2 in sec

Calculation:

From the spring constant graph we have

m = ……gm

x = …….cm

$\therefore$ k = $\frac{mg}{x}$ dyne/cm

From the effective mass curve we have

m' = ……..gm

Result: The spring constant or force constant of the spring is ……dyne/cm and the effective mass of the loaded spring is …….gm. Standard value is 4.9×10^4 dyne/cm^2.

Percent Error: $\frac{Standard\ value \sim Calculated\ value}{Standard\ value} \times 100 = \cdots \%$

Precautions:

1. Loading and unloading of weight must be done gently.
2. Reading should be noted only when tip of pointer comes to rest.
3. Pointer tip should not touch the scale surface.
4. Loading should not be beyond elastic limit.

No. of Experiment- 04

Name of the experiment: Determination of the speed of sound in air by resonance tube method.

Theory: The frequency of sound in a medium is related to the wavelength by the formula, $v = f\lambda$, where v is the speed of sound in the medium, f is the frequency of the sound, and λ represents the wavelength of the sound in the medium. This formula is used to calculate the speed of sound in air from measurements of wavelength and frequency.

The sound speed varies with temperature. At 0℃ the speed in air is 331.4 m/s, while at 20℃ it is 344 m/s. The speed is very nearly linearly dependent on temperature.

When a vibrating object sets up air vibrations in an enclosed space, the sound vibrations in the air are very weak at some frequencies, and strong at other frequencies. The frequencies at which the sound vibrations are strong are called resonant frequencies of the system, and these are easily recognized by listening to the sound intensity.

When the air within an enclosure is set into periodic vibration is set into periodic vibration, the values of many measurable quantities change periodically with time. Such quantities as particle position, velocity, pressure, density, and even temperature vary periodically. The average size of the variation of any one quantity is different in different parts of the medium. There may be certain points in the medium, where a particular quantity is not varying at all. Such a region is called a node of that quantity. The region where the variation of a quantity has a relative maximum is called an antinode of that quantity.

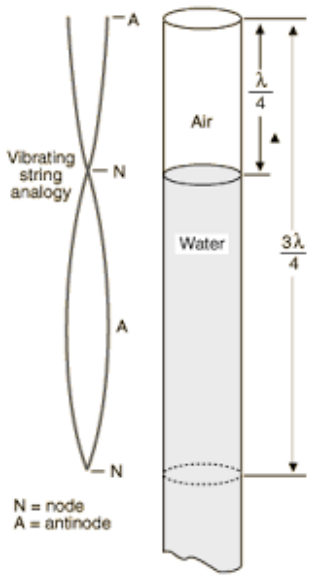


Fig. Vibration of air in resonance tube

In this experiment a tuning fork is used to set up a sound wave of fixed wavelength within the tube. The tube length is then varied until sound resonance occurs at that wavelength. Fig. 2 shows severe resonant situations with a fixed wavelength. The shortest length which produces resonance is approximately one-fourth of the wavelength. Fig.2 shows that the resonant tube lengths differ by one-half wavelength, so if several resonant lengths are determined, their differences give a measurement $\lambda/2$, and from this the speed of sound may be calculated.
i.e.

$$l = \lambda/2 \text{ or } \lambda = 2l$$

So

$$v = f\lambda = 2fl$$

Required Apparatus:

1. Tuning forks
2. Resonance tube
3. Water
4. Rubber hammers, etc.

Procedure:

1. Fill the tube nearly full of water. Strike one of the tuning forks with the rubber mallet supplied and hold it above the water column. Caution: do not touch the tube with the tuning fork - the rapidly moving fork can break the plastic.
2. Using the moveable water reservoir, lower the water surface slowly, listening for amplification of the tone. When a resonance is found, a pronounced reinforcement of the sound will be heard. Move the water surface up and down several times to locate the point of maximum sound intensity and mark that point with a rubber band on the outside of the tube.
3. Lower the water further to find the next resonant length. Continue in this manner as far as the length of the tube will permit. Obtain the lengths $\lambda/4$, $3\lambda/4$, etc. in meters from your measurements. You will need to check to see if your column lengths follow the progression 1, 3, 5, 7, -- since you may have missed a resonance or counted one of the fainter spurious resonances which sometimes occur. Calculate the wavelength and velocity of sound.
4. Repeat the procedure for the other tuning forks supplied. The velocity in miles per hour may be found by multiplying the velocity in m/sec by the factor 2.24. Please record the room temperature for reference since the velocity of sound increases with increasing air temperature.

Observation Table:

Table for speed of sound in air

Frequency, f of the fork (Hz)	**No. of obs.**	**1st resonant length, l_1 (cm)**	**2nd resonant length, l_2 in cm**	**Difference, $l = l_2 - l_1$ cm**	**Speed, $v = 2\,fl$ cms^{-1}**	**Mean, $V = 2fl$ cms^{-1}**

Calculation: At temperature t ℃, v_1 = **….** ms^{-1}

$\mathbf{v}_2$ = **….** ms^{-1}

Average, $v = \frac{V_1+V_2}{2}$ ms^{-1}

At temperature 0 ℃ the velocity is, $V_0 = (v - t\times0.6)$ ms^{-1}

Result: The speed of sound in air at temperature 0℃ is V_0 = ……. ms^{-1}. Standard value is $331.4\ ms^{-1}$.

Percent Error: $\frac{Standard\ value \sim Calculated\ value}{Standard\ value} \times 100 = \cdots\%$

Precautions:

1. **Resonance tube** should be vertical.
2. Pinch cock should be tight.
3. Tuning fork should be vibrated gently by a rubber pad.
4. Prongs should be vibrated in a vertical plane above the mouth (end) of the metallic **tube**.
5. Prongs should not touch the end of the metallic **tube.**

No. of Experiment- 05

Name of the experiment: Determination of the wavelength of monochromatic light (Na) by Newton's rings.

Theory :

NEWTON'S RING

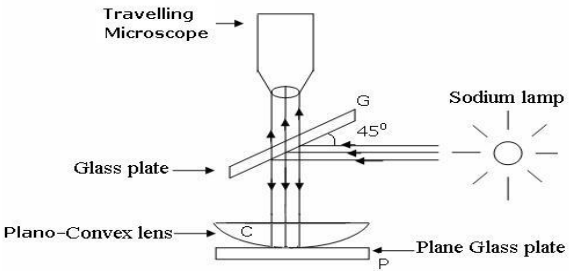


Fig. 1a

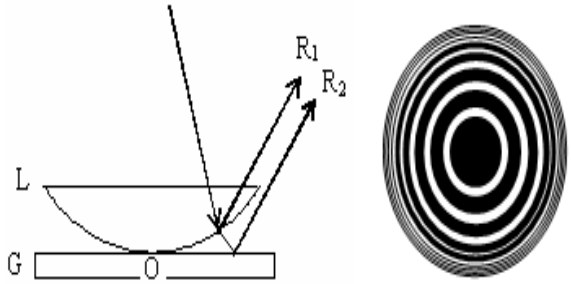


Fig. 1b

Sodium lamp is a source of monochromatic light at the focus of the lens L1. A horizontal beam of light falls on the glass plate G at an angle 45°. The glass plate reflects a part of the incident light towards the air film enclosed by the lens L2 and the glass plate C. The reflected beams from the air film is viewed with a microscope M.

Interference takes place and dark as well as bright circular rings are produced from the lower surface of the lens and upper surface of the glass plate.

Radius of the dark ring:

$$r_n = \sqrt{n\lambda R}$$

This is the radius equation of nth order dark ring.

Radius of the bright ring:

$$r_n = \sqrt{(2n+1)\frac{\lambda R}{2}}$$

This is known as radius eq^n of nth order bright ring.

Now,

Let the diameter of the nth dark ring = D_n

$$r_n{}^2 = n\lambda R$$

But $r_n = \frac{D_n}{2}$

Or $$D_n{}^2 = 4n\lambda R$$

Let the diameter of the $(n+m)^{th}$ dark ring = D_{n+m}

$$r_{n+m}{}^2 = (n+m)\lambda R$$

But $r_{n+m} = \frac{D_{n+m}}{2}$

Or $$D_{n+m}{}^2 = 4(n+m)\lambda R$$

Subtracting equation (ii) & (i) we obtain,

$$\lambda = \frac{D_{n+m}{}^2 - D_n{}^2}{4mR}$$

Where, m = (n+m)-n

Required Apparatus:

1. Two convex lenses
2. Glass plate
3. Na lamp
4. Travelling microscope, etc.

Procedure:

1. If a point source is used only then we require a convex lens otherwise using an extended source, the convex lens L_1 is not required.
2. Before starting the experiment, the glass plates G_1 and G_2 and the plano convex lens should be thoroughly cleaned.
3. The centre of lens L_2 is well illuminated by adjusting the inclination of glass plate G_1 at 45°.
4. Focus the eyepiece on the cross-wire and move the microscope in the vertical plane by means of rack and pinion arrangement till the rings are quite distinct. Clamp the microscope in the vertical side.
5. According to the theory, the centre of the interference fringes should be dark but sometime the centre appears white. This is due to the presence of dust particles between glass plate G_2 and plano-convex lens L_2. In this case the lens should be again cleaned.
6. Move the microscope in a horizontal direction to one side of the fringes. Fix up the crosswire tangential to the ring and note the reading. Again the microscope is moved in the horizontal plane and the cross wire is fixed tangentially to the successive bright fringes noting the vernier readings till the other side is reached.

Observation Table:

Least count, L.C. = 0.01 mm

Table for ring diameter

Ring number	Readings of the microscope								Diameter of the ring D = [L~R] (mm)	D^2 (mm)
	Left side				Right side					
	Linear Scale Reading (S)	Circular Scale Reading, V = [C.S.D × L.C.]	Total reading, L = S+V mm	Mean (L) mm	Main Scale Reading (S)	Vernier Reading, V = [C.S.D × L.C.]	Total reading, L = S+V mm	Mean (L) mm		

Calculation:

Radius of curvature of the plano-convex lens, R = 866.5 mm.

$\lambda_1 = \frac{(D_{n+m})^2 - (D_n)^2}{4mR}$ for ring nth & (n+m)th

$\lambda_2 = \frac{(D_{n+m})^2 - (D_n)^2}{4mR}$ for ring nth & (n+m)th

$\lambda_3 = \frac{(D_{n+m})^2 - (D_n)^2}{4mR}$ for ring nth & (n+m)th

Mean, $\lambda = \frac{\lambda_1 + \lambda_2 + \lambda_3}{3}$ m

Result: The wavelength of monochromatic light (Na) is λ = …… Standard value is 5893 $\overset{0}{A}$.

Percent Error: $\frac{Standard\ value \sim Calculated\ value}{Standard\ value} \times 100 = \cdots \%$

Precautions:

(i) The intensity of the ring system decreases as one goes from the inner to the outer rings, thus setting a limit for the selection of the outermost ring whose diameter is to be determined.

(ii) Newton's rings can also be observed in transmitted light but in that case the rings will be less clearly defined and less suited for measurement.

(iii) The first few rings near the centre may be deformed due to various reasons. The measurement of diameters of these rings may be avoided.

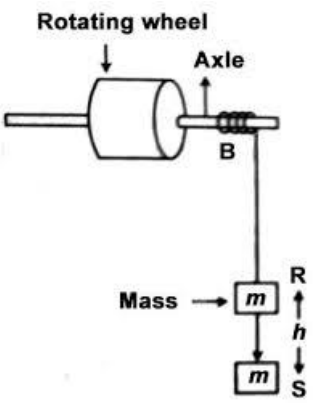
(iv) For the purpose of the experiment a convex lens whose radius of curvature is of the order of 100 cm is suitable. Otherwise the diameters of the rings will be too small and it will be difficult to measure them.

(v) It may be noticed that the inner rings are somewhat broader than the outer ones.

No. of Experiment- 06

Name of the experiment: Determination of the moment of inertia of a flywheel about its axis of rotation.

Theory: Flywheel resists changes in rotational speed by its moment of inertia. A torque aligned with its axis of symmetry. Mechanical energy is stored in a flywheel.



Mathematical expression: Moment of inertia of a body rotating about an axis of rotation is $I = Mr^2$.

Uses: Flywheels are used to provide continuous power output in system where the energy source is not continuous. (rice machine, water supply machine, Akh mara machine) Industrial flywheel, tractor with exposed flywheel, automobile engine flywheel, tight rope walkers use the moment of inertia of a long rod for balance as they walk the rope.

Conservation of energy:

Potential energy of mass, m = K.E. of m + K.E. of flywheel + Work done against friction

$$mgh = \frac{1}{2} m\omega^2 r^2 + \frac{1}{2} I\omega^2 + n_1 F \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(i)$$

But work done by frictional force,

$$Fn_2 = \frac{1}{2} I\omega^2$$

Or,

$$F = \frac{\frac{1}{2} I\omega}{n_2}$$

From equation (i) we get,

$$mgh = \frac{1}{2} m\omega^2 r^2 + \frac{1}{2} I\omega^2 + n_1 . \frac{\frac{1}{2} I\omega^2}{n_2}$$

$$I = \frac{2mgh - m\omega^2 r^2}{\omega^2 (1 + \frac{n_1}{n_2})}$$

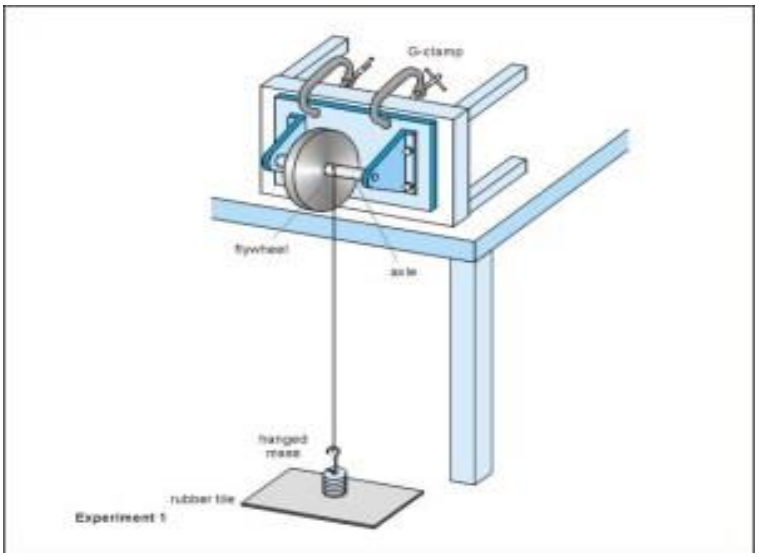
This is the equation for moment of inertia of the flywheel.

Where, $\omega = \frac{4\pi n_2}{t}$.

<u>Required apparatus:</u>

1. Flywheel,
2. String,
3. Stop-watch
4. Vernier calipers
5. Meter ruler,
6. slotted mass on hanger, etc.

Procedure:



1. The flywheel is set with the axle of the flywheel horizontal. A polystyrene tile was placed on the floor to avoid the impact of the mass on the floor.
2. The vernier caliper is used to measure the diameter d of the axle. The mean of two perpendicular measurements was taken.
3. The hanger with appropriate amount of slotted mass is put on the tile. Which is used the balance to measure the total mass m.
4. The mass is winded to an appropriate height. Verified that the string fell off the axle when the mass hit the ground.
5. The height h of the mass is measured. The height h was recorded.
6. The mass is released then and at the same time the stop-watch is started.
7. As soon as the mass hit the ground, timing was stopped and the number of revolution the wheel revolved is taken.
8. The mass is winded up again and steps are repeated for at least 3 times.The mean time of the falling is obtained.

Observation Table:

Table for the determination of n_1, n_2 and t

No. of Obs.	Mass m in gm	Height h in cm	No. of revolutions n_1	Mean n_1	No. of revolutions n_2	Mean n_2	Time t (sec)	Mean t (sec)

Calculation:

For the load, m_1 = ……gm

h_1 = ……cm

n_1 = …..

n_2 = …..

r = ……..cm

$$\omega_1 = \frac{4\pi n_2}{t}$$

So

$$I_1 = \frac{2m_1gh_1 - m_1{\omega_1}^2r^2}{{\omega_1}^2(1+\frac{n_1}{n_2})}$$

Similarly, for the load m_2 = ……gm

h_2 = ……cm

n_1 = …..

n_2 = …..

r = ……..cm

$$\omega_2 = \frac{4\pi n_2}{t}$$

$$\therefore\ I_2 = \frac{2m_2gh_2 - m_2{\omega_2}^2r^2}{{\omega_2}^2(1+\frac{n_1}{n_2})}$$

Mean value, $I = \frac{I_1 + I_2}{2}$ gm.cm^2

Result: The moment of inertia of the flywheel is …….gm.cm^2.

Standard value is 2.5×10^5 gm.cm^2.

Percent Error: $\frac{Standard\ value \sim Calculated\ value}{Standard\ value} \times 100 = \cdots\%$

Precautions:

1. Do not stand too close to the polystyrene tile when releasing the mass.
2. The turns of the string must not overlap the other.
3. Stop Watch must be handled with care in order to avoid any error in the reading.
4. The first few turns of the string should overlap the others.
5. The mass should be wound up to the same height in all trials.

No. of Experiment- 07

Experiment Name: To determine the value of 'g' acceleration due to gravity by means of compound pendulum.

Theory: Compound pendulum is a rigid body of any shape free to turn about a horizontal axis. When the body oscillates about a hole from one end or the corresponding hole from the other end, then the time periods are the same. Then the time period of oscillation T is found from the following relation,

$$T = 2\pi\sqrt{\frac{L}{g}}$$

$$or, g = 4\pi^2\frac{L}{T^2}$$

By finding L graphically, and determining the value of the period T, the acceleration due to gravity 'g' at the place of the experiment can be determined.

Fig.1

If a uniform rod is used as a compound and T is measured for different values of l on either side of the C.G., then a graph like below may be obtained.

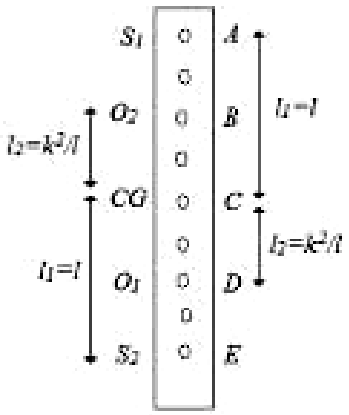


Fig.1

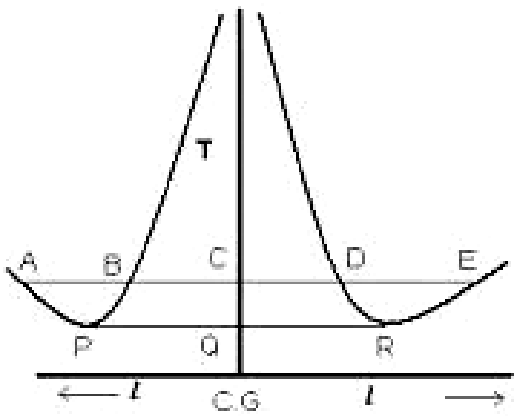


Fig. 2

From graph, we will get T and L and hence, $g = 4\pi^2\frac{L}{T^2}$ will be got.

Apparatus: A set of compound pendulum, stop watch, meter scale, etc.

Procedure: 1) Find out the center of gravity G of the bar.
2) Insert a metal wedge in the first hole in the bar towards A and place the wedge on the support S_1S_2 so that the bar can turn round S.
3) Set the bar to oscillate taking care to see that the amplitude of oscillations is not more than 5_o. Note the time for 20 oscillations.
4) Measure the length from the end A of the bar to the top of the first hole i.e., up to the point of suspension of the pendulum.
5) In the same way, suspend the bar at holes 2, 3, …… and each time note times for 20 oscillations. Also measure distances from the end A for each hole.
6) When the middle of the bar is passed, it will turn round so that the end B is now on the top. But continue measuring the distances from the point of suspension to the end A.
7) Now calculate the time period T from the time recorded for 20 oscillations.
8) On a nice and large graph paper, plot a curve with length as abscissa and period T as ordinate with the origin at the middle of the paper along the abscissa.

Observation Table:

Table-1 for time period

At the top	Hole No.	Distance from G in cm	Time for 20 oscillations, t (s)	Time period $T = \frac{t}{20}$ s
End A	2			
	3			
	4			
	5			
	6			
	7			
	8			
End B	2			
	3			
	4			
	5			
	6			
	7			
	8			

Calculation:

From the graph, AC = …… cm

BD = …… cm

Mean length, $L = \frac{AC+BD}{2}$ = ….. cm

Corresponding time period T = ….. second

Acceleration due to gravity,

$$g = 4\pi^2 \frac{L}{T^2}$$

= ….. cm/sec^2

= ….. m/s^2

Result: Acceleration due to gravity, g = ….. m/s^2. Standard value is 9.8 ms^{-2}.

Percent Error: $\frac{Standard\ value \sim Calculated\ value}{Standard\ value} \times 100 = \cdots \%$

Precautions:

(i) Distances are to be measured from the point G or the top.

(ii) In measuring time an accurate stop-watch should be used.

(iii) Oscillations should be counted whenever the line of the bar crosses the intersecting point of the cross-wires, in the same direction.

(iv) Graph paper used should have sharp lines and accurate squares and should be sufficiently large to draw smooth and large curves.

(v) Amplitude of oscillations must not be more than 5°.

No. of Experiment- 08

Name of the Experiment: Determination of the specific **resistance** (resistivity) of a wire by Meter Bridge.

Theory: The amount of resistance or hindrance that a conductor having unit length and the unit area offers to the flow of current is called specific resistance of that conductor. It is denoted by ρ.

If the length of a wire is L, area of cross-section is A and total resistance is Q, then its specific resistance is –

$$\rho = QA/L$$

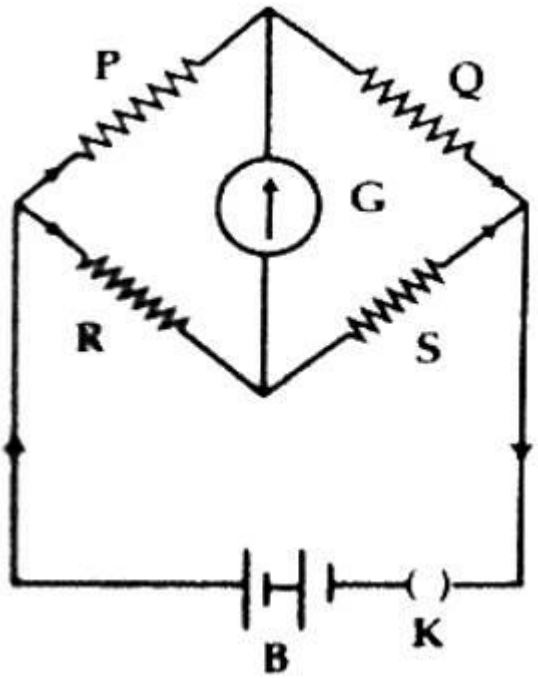
If the wire is cylindrical or circular then cross-section, $A = \pi r^2$.

Here, r = radius of the wire.

Specific resistance, $\rho = Q\pi r^2 / L$ … … … (1)

If R is expressed in the practical unit and L and r are expressed in meter, then the unit of ρ will be ohm-meter (Ω-m).

The total resistance Q in the equation (1) can be determined by a meter bridge according to the following theory.



In the left of the two gaps of the meter bridge a known resistance P and in the right gap an unknown resistance Q is inserted [Figure] and after completing the circuit if the distance of the null point is l from the left terminal and if σ is the resistance of the unit length of the wire, then applying Wheatstone principle we get,

P/Q = R/S = [Resistance of length l of the bridge wire / Resistance length (100 – l) of the bridge wire]

or, $$P/Q = R/S = l\sigma / (100 - l)\sigma = l / (100 - l)$$

or, $$Q = [P \times (100 - l)] / l \text{ ohm} \ldots \ldots \ldots (2)$$

Again, by changing the position of the resistances i.e., known resistance P in the right gap and unknown resistance Q in the left gap, we get,

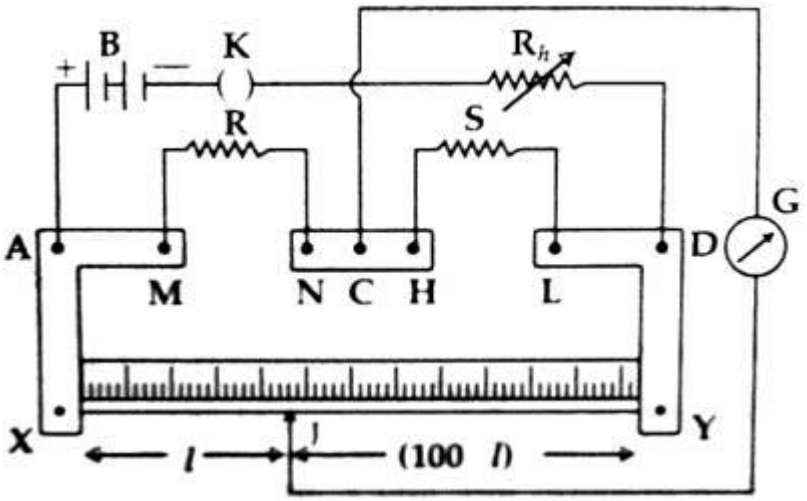
$$Q/P = l / (100 - l)$$

or, $$Q = P \times l / (100 - l) \text{ ohm} \ldots \ldots \ldots (3)$$

From the average value of the equations (2) and (3), we can find out the unknown resistance Q.

By measuring the radius r, length L of the wire and by determining the unknown resistance by the meter bridge and then inserting these values in equation (3) we can find out the value of the specific resistance of the wire.

Description of the wire: It consists of three thick copper or brass strips XAM, NCH and LDY mounted on a long wooden board [Figure-2]. The resistances of these strips are negligible and there is connecting screw at each point of A, M, N, C, H, L, and D. There is a meter scale placed between the strips XAM and LDY. A straight wire of lm long and of uniform cross-section is attached at points X and Y along the length of the scale. A 1-meter long wire is used as a resistor; hence it is called a meter bridge.



<u>Required instrument:</u> (1) A meter bridge, (2) electric cell, (3) galvanometer, (4) resistance box. (5) commutator or key, (6) jockey, (7) connecting wires, (8) screw gauge, (9) meter scale, (10) experimental wire etc.

<u>Procedure:</u>

1. The circuit is drawn in a rough notebook.
2. The connection is made according to the circuit diagram.
3. The unknown resistance Q is connected to the right gap and the known resistance P is connected to the left gap of the bridge. The galvanometer is connected from the midpoint of the meter bridge and through the resistance box to the jockey J or a push key.
4. After completing the circuit connecting the jockey is touched at one end of the wire of the bridge and the deflection of the galvanometer is observed. Again the jockey is touched at the other end of the meter bridge and deflection is noticed. If these two deflections are found opposite then it is considered that the circuit connection is alright.
5. A known resistance P_1 is applied in the circuit from the resistance box. Now holding the jockey on the wire it is moved right and left and a position on the wire is fixed where at the galvanometer does not show any deflection. This is the null point. The length l_1 is determined from the left of the meter scale attached to the bridge. Inserting these values in equation (2) the value of Q is found out.
6. Now positions of P and Q are interchanged. Keeping the value of P_1 fixed following the above procedure the null point is found out and the length l_2 up to that point is noted. These values are inserted in equation (3) and Q_1 is found out.
7. By interchanging the positions of P and Q are taken to the earlier position. A resistance P_2 of different value is withdrawn from the box P and following procedures (5) and (6) unknown resistances Q_2 and Q_2‘ are found out.

8. Following the procedure (7) resistance P_3 is taken out from the resistance box P and following the previous procedures Q_3 and Q_3'. are found out.
9. Now taking the average values of Q_1, Q_1‘, Q_2, Q_2‘ and Q_3 and Q_3‘ the unknown resistance Q ohm is found out.
10. By removing the experimental wire from the meter bridge its average length is measured by a meter scale.
11. Diameter i.e., the radius of these wire is measured by a screw gauge and cross-sectional area of the wire is determined.
12. Inserting the experimental values of Q, L and r in equation (2), the specific resistance of the wire is calculated.

Observation Table:

Readings for resistance of the wire

No. of observations	**Resistance P in Ohms**	**Reading of null point**		**Unknown resistance** $Q = P \times \frac{(100-l)}{l}$ **Ohms**	**Mean Q Ohms**
		l	**100-l**		
1	7				
2	10				
3	13				
4	16				

Calculation:

(A)Length of the wire (L)

(i).....cm,(ii).....cm,(iii).....cm.

Mean L=.....cm

(B) Diameter of the wire (d).

L.C. of the screw-gauge=.....cm

(i) d_1=.....cm,(ii) d_2=.....cm,(iii) d_3=.....cm

Mean diameter of the wire, d=.....cm.

Radius of the wire,

r=d/2....cm

Specific resistance of the material of the given wire is,

$$\rho = Q\pi r^2 / L$$

Results: Specific resistance of the material of the given wire is..... (Ω-m). Standard value is 2×10^{-5} ohm-cm.

Percent Error: $\frac{Standard\ value \sim Calculated\ value}{Standard\ value} \times 100 = \cdots \%$

Precautions:

(i) See that the null point is not far away from the middle.

(ii) It is essential to see that none of the plugs in the resistance box R is loose.

(iii) Take care to determine the diameter (d) of the wire very accurately.

(iv) On reversing the current if the null point changes appreciably, the thermo-electric effect is too large. In such a case close the galvanometer circuit keeping the battery circuit open. The deflection of the galvanometer should be taken as the zero when looking for a null point.

(v) E.M.F. of the cell should be checked before starting the experiment.

No. of Experiment- 09

Name of the experiment: Determination of the charge mass ratio (e/m) of an electron by Helmholtz Coil.

Theory: Charged particles in a magnetic field feel a force (magnetic force) perpendicular to their velocity. Since their movement is always perpendicular to the force, magnetic forces due no work and the particle's velocity stays constant. Since the force is $F_m = evB \sin\theta$ in a constant magnetic field (B), a charged particle feels a force of constant magnitude always directed perpendicular to its motion. The result is a circular orbit.

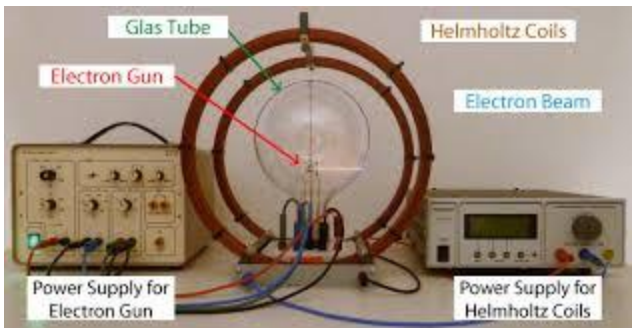
The radius of the orbit depends on the charge and velocity of the particle as well as the strength of the magnetic field. The acceleration of a particle in a circular orbit is:

$a = \frac{v^2}{R}$, centripetal force is $F_c = \frac{mv^2}{R}$, i.e. $F_c = F_m$

Using F = ma

$\therefore\ F = evB = m\frac{v^2}{R}$

Or, $R = \frac{mv}{eB}$



Thus the radius of the orbit depends on the particle's momentum 'mv' and the product of the charge and strength of the magnetic field.

Now, $v = \frac{eBr}{m}$

If the electron is initially accelerated by an applied potential difference, V, then we have

An electron with charge e and ,mass m, when accelerated through a potential difference V gains a kinetic energy $\frac{1}{2}mv^2$.

Electron volt is defined as the amount of energy an electron gains after being accelerated by a potential difference. This energy is equal to the work done on an electron in accelerating it.

i.e. Kinetic energy = Electron Volt

$$\frac{1}{2}mv^2 = eV$$

[(i) if $\theta<90^0$, path is helical; (ii) if $\theta=90^0$, path is circular]

$$mv^2 = 2\ eV$$

$$m\left(\frac{eBr}{m}\right)^2 = 2eV$$

$$\frac{e}{m} = \frac{2V}{B^2r^2}$$

The required B is set up by a Helmholtz coil which produces a fairly uniform B near its centre is

$$B = \mu_0\left(\frac{4}{5}\right)^{\frac{3}{2}}\frac{ni}{a} \quad \text{.....(i)}$$

Here, n = number of turns in each coil = 124
a = radius of the coil = 147.5 mm
i = current in the coil
μ_0 = permeability constant = $4\pi\times10^{-7}$ (wb-m)/A

N.B. Since the magnetic force is always perpendicular to the velocity of a charged particle, the particle will undergo circular motion.

Required apparatus:

1. Electron beam,
2. Helmholtz coils
3. Power supply, etc.

Observation Table:

Table for radius of the path of electron

Current i (amp)	**voltage V (volts)**	$B = \mu_0\left(\frac{4}{5}\right)^{3/2}\frac{ni}{a}$	**Diameter of beam path, D (cm)**	**Radius r (m)**	**Mean r (m)**	**r^2 (m)**

Calculation:

1. e/m =
2. e/m =
……..
Mean, e/m =

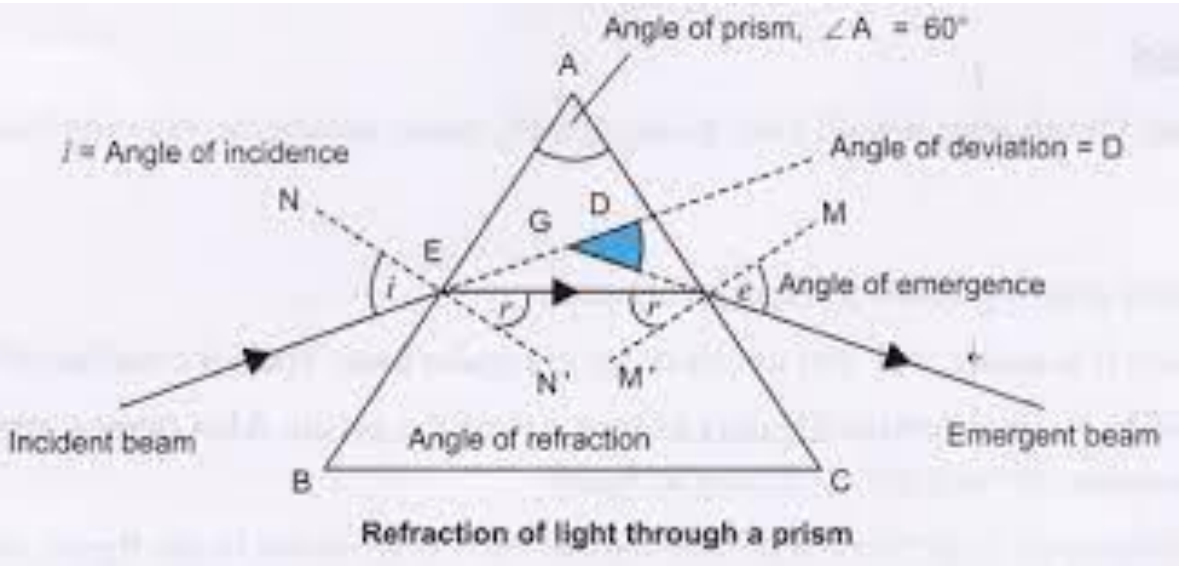
Result: The specific charge of electron is ….. Standard value is 1.75×10^{11} C/Kg.

Percent Error: $\frac{Standard\ value \sim Calculated\ value}{Standard\ value} \times 100 = \cdots \%$

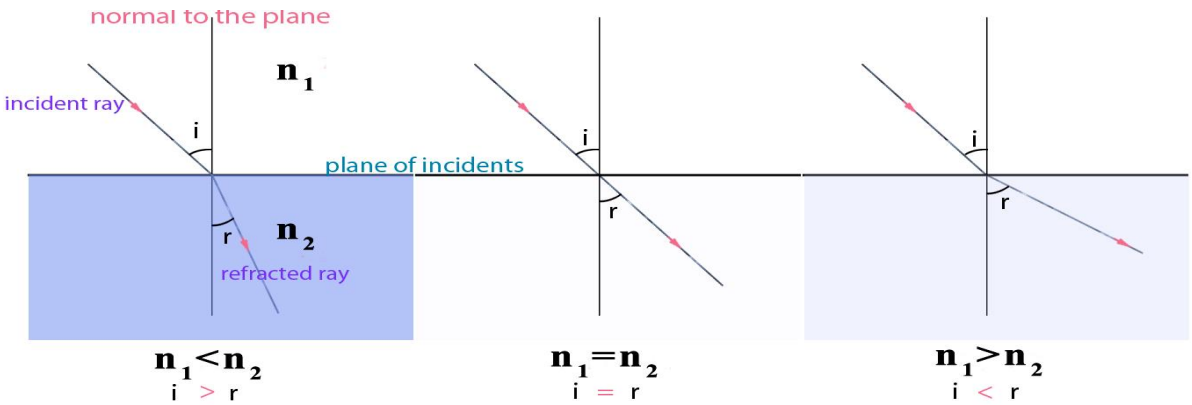
Precautions:
(i) The value of e/m is a universal constant and is independent of the method by which it is determined, the type of gas tube used, the material of which the cathode is made, etc.
(ii) Care should be taken to determine B as accurately as possible.
(iii) The accuracy of the experiment depends on the degree of accuracy with which B is determined as B appears in second power in the expression for e/m. It is assumed in this experiment that the value of B is constant throughout the entire path of the electron.
(iv) The path of the electrons should be made as circular as possible. It is then that the relation F = ev×B is valid.
(v) The value of e/m as obtained from very accurate experiments is 1.76×10^{11} C/Kg.

No. of Experiment- 10
Name of the experiment: Determination of the refractive index of the material of a prism.

Theory:

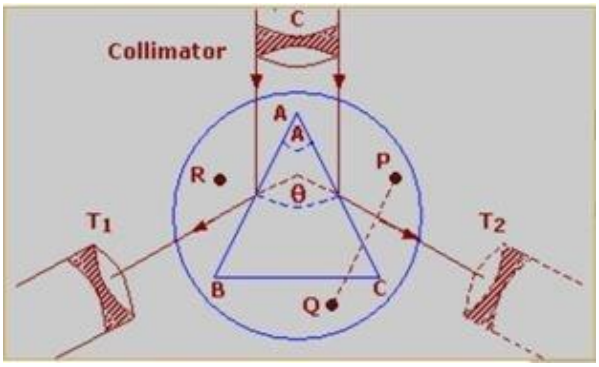


Refraction of light through a prism



The above figure illustrate the change in refracted angle with respect to the refractive index,

Angle of prism: A prism has 6 angles of prism or vertices. The opposite side to the angle of prism is the base of the prism.
Here, AB & AC are two refracting surfaces of the prism, A is the angle of prism, BC is the base of the prism and ABC is the section of the prism.
When a beam of parallel light falls on the edge of a prism, light is reflected partly from one and partly from the other refracting surface of the prism & the angle between the reflected rays is twice the angle of the prism.



Figure

Let PQ and P'Q' are two parallel rays falling on the two different faces (AB & AC) of the prism.
QR and Q'R' are the reflected rays of PQ and P'Q'. When the reflected rays produced let them meet backward to meet at point D.
Hence, $\angle D = 2\angle A$

$$\angle A = \frac{\angle D}{2}$$

$$\therefore \quad A = \frac{D}{2}$$

Where, A is the angle of prism.

Refractive index: If A is the angle of a prism and δ_m is the minimum deviation for a given colour or wavelength, then the refractive index of the material of the prism is

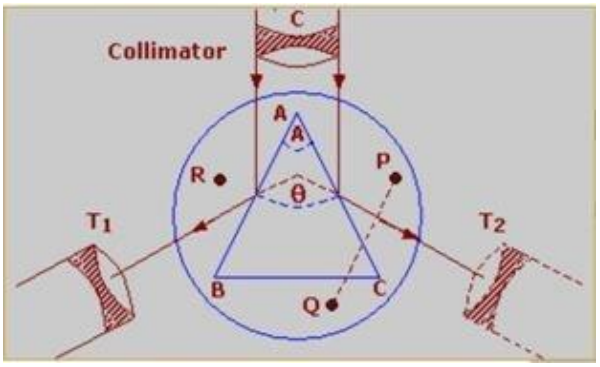
$$\mu = \frac{\sin(\frac{A+\delta_m}{2})}{\sin\frac{A}{2}}$$

Where, δ_m is the angle of minimum deviation.

Angle of deviation (δ): The angle bet^n the ray of incidence and the ray of emergence is called the angle of deviation. If I is extended, δ decreases.

Angle of min^m deviation (δ_m): The angle of deviation becomes min^m for a particular angle of incidence. This min^m value of the angle of deviation is called the angle of min^m deviation (δ_m). The position of the prism for which the angle of min^m deviation is found, that position (I = r) is called min^m deviation position.

If i is increased, then δ will be min^m at a particular i and in that position δ does not decrease but increase. Then δ is δ_m.



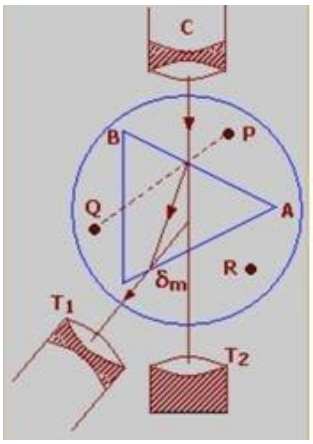


Figure: Left: Arrangement to determine the angle of prism
Right: Arrangement to determine the angle of minimum deviation.

Required Apparatus:

1. Spectrometer
2. Prism
3. Travelling telescope,
4. Collimator
5. Prism table, etc.

Observation Table:

(i) Value of the one division of the main scale = ……
degrees Total number of vernier divisions = …….

Least count of the vernier = ……. degrees = …… second

Table for the angle of minimum deviation (δm)

Vernier number	No. of observations	Reading for the minimum deviation position					Reading for the direct position					Angle of minimum deviation (δm)	Mean δm
		Main scale reading (S)	Vernier scale divn. (V.D)	Vernier scale reading V = V.D×V.C.	Total reading M = S+V	Mean M	Main scale reading (S)	Vernier scale divn. (V.D)	Vernier scale reading V = V.D×V.C.	Total reading N = S+V	Mean N		
I													
II													

Calculation:

Angle of the prism = 60°
Angle of minimum deviation for yellow = ……..
Refractive index for yellow = …….

Result: Refractive index for the material of the prism. Standard value is 1.53.

Percent Error: $\frac{Standard\ value \sim Calculated\ value}{Standard\ value} \times 100 = \cdots \%$

Precautions:

(i) The source must be in front of the collimator slit so that the image appears bright.
(ii) The vertical cross-wire or better the centre of the cross-wires should be made coincident with the same edge of the slit image.
(iii) Care should be taken so that there is no parallax between the cross-wires and the slit image.
(iv) For the final setting, the telescope must be rotated carefully with the tangent screw always in the same direction so as to avoid back-lash error.
(v) While measuring the angle of the prism, the vertex of the prism should be placed at the center of the table.

No. of Experiment- 11

Name of the experiment: Determination of the specific heat of a solid by the mixture method.

Theory:

Calorimetry: When two bodies at different temperatures are put in contact, then the heat flows from hot body to cold body.

If no heat enters or leaves through the system when the two bodies are in contact and if no chemical action takes places between them, then

Heat lost by one quantity = Heat gained by other quantities

Specific heat (S): The quantity of heat necessary for a body of unit mass to raise unit temperature is called specific heat.

$$\therefore \qquad S = \frac{dQ}{md\theta}$$

Where, S = specific heat of an object
dQ = absorbed/rejected heat
m = mass of the object
dθ = difference of temperature

Required apparatus:
1. Balance
2. Steam boiler
3. A piece of the solid
4. thermometers, etc.

Observation Data:
Let the mass of the solid object = M gm
Specific heat of the solid = S cal/gm.℃
Temperature of the heated object = θ_1℃ = 60-65℃
Mass of the beaker = m_1 gm
Specific heat of the beaker = s_1 cal/gm.℃= 0.12 cal/gm.℃
Mass of the water in the beaker = m_2 gm
Specific heat of the water = s_2 cal/gm.℃ = 1 cal/gm.℃
Temperature of beaker and water = θ_2℃
Temperature of the mixture = θ℃ (Temperature at which the solid and water are at equilibrium)

Calculation:
According to the principle of calorimetry,
Heat lost by the solid = Heat gained by beaker + Heat gained by water

$$MS\,(\theta_1-\theta) = m_1s_1\,(\theta-\theta_2) + m_2s_2\,(\theta-\theta_2)$$
$$= (m_1s_1 + m_2s_2)(\theta - \theta_2)$$

$$S = \frac{(m_1 s_1 + m_2 s_2)(\theta - \theta 2)}{M\ (\theta 1 - \theta)}$$

This is the equation for the determination of the specific heat of the solid body.

Result: The specific heat of the solid object is S =... cal/gm.℃. Standard value is 0.092 $cal/gm.$℃ for brass.

Percent Error: $\frac{Standard\ value \sim Calculated\ value}{Standard\ value} \times 100 = \cdots \%$

Precautions :

(i) The solid used should not be too small so that the percentage error due to loss of heat during the transfer of the hot solid is reduced.

(ii) The quantity of water used should be such that it just completely covers the body, so that the total rise of temperature is not very small (about 5℃).

(iii) The solid should be quickly transferred to the calorimeter so that hardly any heat is lost during the process.

(iv) The calorimeter vessel should be suitably screened from any extraneous heating effect.

No. of Experiment- 12

Name of the experiment: Determination the value of J, the mechanical equivalent of heat, by electrical method.

Theory: The mechanical equivalent of heat J is the number of Joules of electrical energy required to generate one calorie of heat.

If E volt be the potential difference across a conducting coil and I ampere be the current flowing through the coil for t seconds, then electrical energy expended in the coil is Eit. Hence if this energy is converted into H calories of heat we shall have

$$J = \frac{Eit}{H} \text{ Joules per calorie} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

If the heat thus developed be measured by means of a calorimeter, then the temperature of the calorimeter with its contents will rise from θ_1℃ to θ_2℃.Hence the heat taken up by the calorimeter and its contents is given by

$$H = (Ms + mS)(\theta_2 - \theta_1) \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

Where, M=mass of the liquid in the calorimeter

s= specific heat of the liquid

mS = water equivalent of the calorimeter and the stirrer.

From (1) and (2), we get

$$J = \frac{Eit}{(Ms + mS)(\theta 2 - \theta 1)} \text{ Joules per calorie}$$

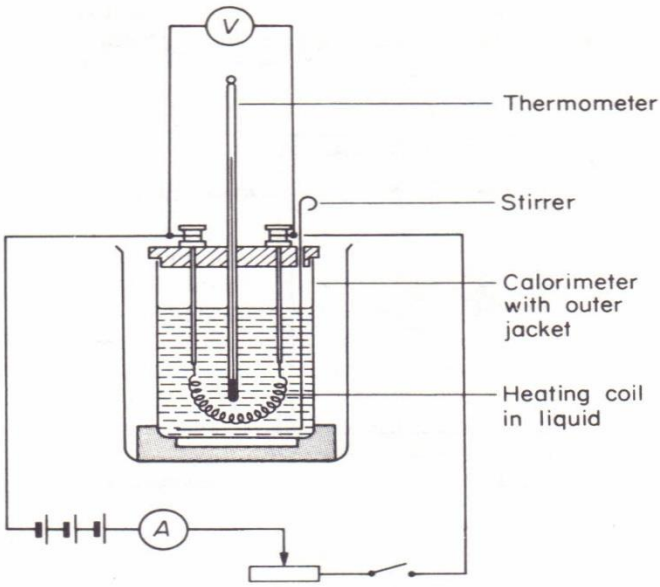


Fig. 42.2. Specific heat capacity of a liquid (electrical method)

Required apparatus:

1. Calorimeter
2. Heating coil
3. Stirrer
4. Thermometers, etc.

Observation Data:

Suppose, mass of the calorimeter =Mg

Mass of the liquid=mg

Specific heat of the calorimeter=s calg^{-1}°C^{-1}

Specific heat of the liquid=S calg^{-1}°C^{-1}

Initial temperature of the calorimeter and the liquid=θ_1°C

Current=i A

Potential difference between the terminal of the coil=V volt

Time of current flow=t second

Maximum temperature of the calorimeter and liquid= θ_2°C

Now heat absorbed by the calorimeter and the liquid,

H=(Ms+mS)(θ_2- θ_1) cal

And work done due to flow of current

$$W = Vit\ J$$

$$\therefore \quad J = \frac{W}{H} = \frac{Vit}{(Ms+mS)(\theta 2 - \theta 1)}\ Jcal^{-1}$$

Now, by knowing the values of V, i, t, M, m, s, S, = θ_1 and = θ_2 the value of J can be found out.

Result: The mechanical equivalent of heat, J=……Jcal^{-1}. Standard value is 4.2 J/cal.

Percent Error: $\frac{Standard\ value \sim Calculated\ value}{Standard\ value} \times 100 = \cdots \%$

Precautions:

1. Wires must be connected properly with power supply, ammeter, voltmeter and calorimeter.
2. Thermometers should not touch the coil.
3. If the rise of temperature is greater than 10℃, then the radiation correction of temperature will be large which would cause errors.

No. of Experiment- 13

Name of the Experiment: To determine the plateau and optimal operating voltage of a Geiger-Muller counter.

Required instrument:

- Radioactive Source (e.g., Cs-137, Sr-90, or Co-60).
- Set-up for ST-360 Counter with GM Tube and stand (Counter box, power supply, GM Tube, shelf stand, serial cable, and a source holder For the stand).

Figure 1: experimental setup

Theory:

Radioactive materials that we find as naturally occurring were created by:

1. Formation of the universe, producing some very long lived radioactive elements, such as Uranium and Thorium.
2. The decay of some of these long lived materials into other radioactive materials like Radium and Radon.
3. Fission products and their progeny (decay products), such as Xenon, Krypton, and Iodine. Man-made radioactive materials are most commonly made as fission products or from the decays of previously radioactive materials. Another method to manufacture radioactive materials is activation of non-radioactive materials when they are bombarded with neutrons, protons, other high energy particles, or high energy electromagnetic waves.

The Geiger-Müller Counter

Geiger-Müller (GM) counters were invented by H. Geiger and E.W. Müller in 1928, and are used to detect radioactive particles (α and β) and rays (γ and X).

A GM tube usually consists of an airtight metal cylinder closed at both ends and filled with a gas that is easily ionized (usually neon, argon, and halogen). One end consists of a "window" which is a thin material, mica, allowing the entrance of alpha particles. (These particles can be shielded easily.) A wire, which runs lengthwise down the center of the tube, is positively charged with a relatively high voltage and acts as an anode. The tube acts as the cathode. The anode and cathode are connected to an electric circuit that maintains the high voltage between them.

When the radiation enters the GM tube, it will ionize some of the atoms of the gas (counting gas).

Due to the large electric field created between the anode and cathode, the resulting positive ions and negative electrons accelerate toward the cathode and anode, respectively. Electrons move or drift through the gas at a speed of about 104 m/s, which is about 104 times faster than the positive ions move.

The electrons are collected a few microseconds after they are created, while the positive ions would take a few milliseconds to travel to the cathode.

As the electrons travel toward the anode they ionize other atoms, which produces a cascade of electrons called gas multiplication or a (Townsend) avalanche. The multiplication factor is typically 106 to 108.

The resulting discharge current causes the voltage between the anode and cathode to drop. The counter (electric circuit) detects this voltage drop and recognizes it as a signal of a particle's presence.

There are additional discharges triggered by UV photons liberated in the ionization process that start avalanches away from the original ionization site.

These discharges are called Geiger-Müller discharges. These do not affect the performance as they are short-lived.

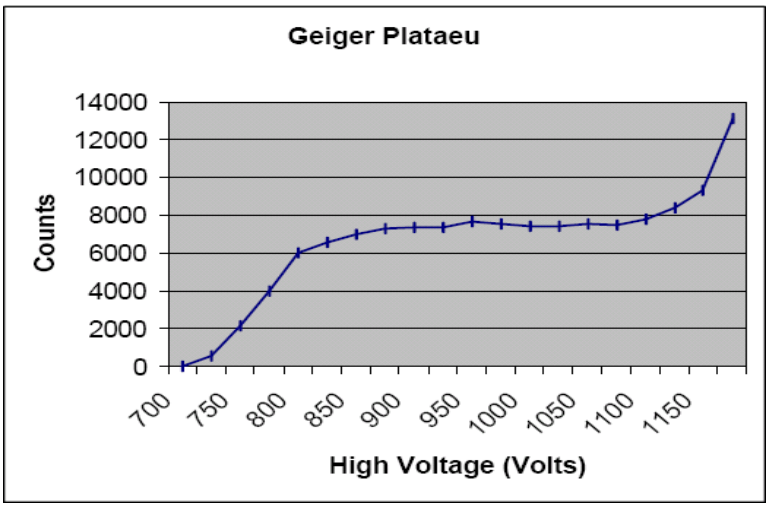


Figure 2: typical GM plateau

By the end of this experiment, you will make a graph similar to the one in Figure 2, which shows a typical plateau shape.

Procedure:

1. Plug in the transformer/power supply into any normal electricity outlet and into the back of the ST-350 box.
2. Remove the red endcap from the GM tube **VERY CAREFULLY**. Place the GM tube into the top of the shelf stand with the window down and BNC connector up.
3. Attach the BNC cable to the GM tube and ST-350.
4. Attach the serial cable to the ST-350 and a serial port on the back of your PC (if you are using one).
5. Turn the power switch on the back of the ST-350 to the ON position, and turn the **DISPLAY FUNCTION** knob on the front face to the remote setting
6. Open the computer interface by pressing the START key on the PC. Then look under Programs -> SpecTech options for the ST350 option. You should then see the blue control panel appear on your screen.
7. Go to the **Setup menu** and select the **HV Setting option**. In the **High Voltage (HV) window**, start with 700 Volts. In the **step voltage window**, enter 20. Under Step Voltage Enable, select **On** (the default selection is off). Finally, select **OK**.
8. Go under the **Preset option** and select **Preset Time**. Enter 30 for the number of seconds and choose **OK**. Then also under the **Preset option** choose **Runs**. In the window, enter 26 for the number of runs to make.
9. You should see a screen with a large window for the number of counts and data for all the runs on the left half of the screen. On the right half, you should see a window for the preset time, elapsed time, runs remaining, and HV and step voltages. If not, go to the view option and select counts.
10. Make sure no other previous data by choosing the erase button (last one on the right). Then select the green diamond to start taking data.
11. When all the runs are taken, choose the File menu and Save As. Then you may save the

data file anywhere on the hard drive or onto a floppy disk. The output file is a text file that is tab delimited, which means that it will load into most spreadsheet programs. See the Data Analysis section for instructions in doing the data analysis in Microsoft Excel ®. Another option is that you may record the data into your own data sheet and graph the data on the included graph paper.

12. You can repeat the data collection again with different values for step voltage and duration of time for counting. However, the GM tubes you are using are not allowed to have more than 1200 V applied to them. Consider this when choosing new values.

Observation Table:

Data Table for Geiger Plateau with for a radioactive sources

No. of Obs.	Source	High Voltage	Counts in 30 s		Mean Counts	Counting rate
			1	2		

Calculation: From graph we have,

V_1 =

V_2 =

Optimal operating voltage = $\frac{V_1+v_2}{2}$

Result: The Optimal operating voltage is…… Volts.

Precautions:

1. Do not let anything touch the window.
2. Do not ever apply a high voltage beyond the plateau region, as the tube will be damaged.
3. Counting rates of more than 100000/min will likely be subject to some coincidence error.

Experiment No- 14

Name of the experiment: To calibrate a polarimeter and hence to determine the specific rotation of a sugar solution by means of a polarimeter.

Theory: When a polarized light beam of λ wavelength passes through optical active material the plane of polarization of the beam is rotated to the θ angle. The rotation angle depends on (i) the concentration and nature of the solution, (ii) the path travelled by the light rays through the solution.

Generally, the angle of rotation of the plane of polarization at temperature t°C is proportional to rotation angle, θ. The concentration of the solution, C and the path length, L in the solution,

$$\theta \propto C \text{ and } \theta \propto L$$

∴ $$\theta \propto LC$$

or, $$\theta = SLC$$

∴ $$S = \frac{\theta}{LC} \quad \text{..... (i)}$$

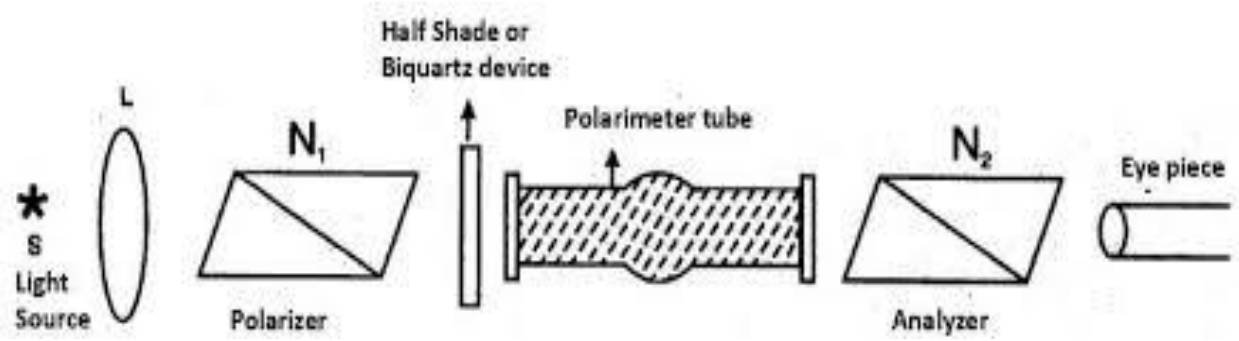
Here, S is a constant. This is called specific rotation.

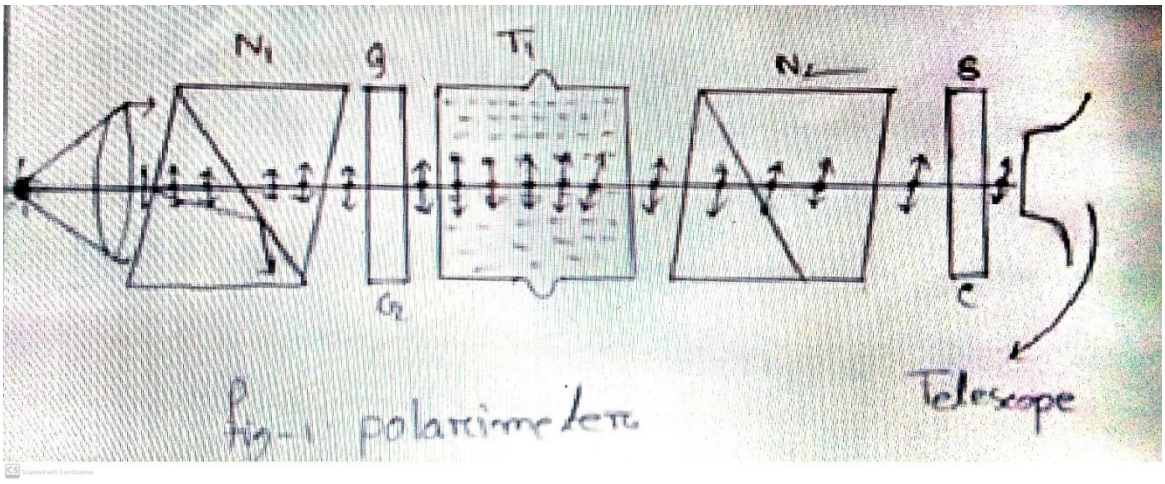
If there is 1 gm of active substance per 1 cc of solution at constant temperature, the rotation of 1 decimeter length of solution is called specific rotation.

Specific rotation refers to *the angle of rotation made by a chiral compound when plane-polarized light is passed* through it at a specific wavelength.

So, from equation (i)

$$S = \frac{10\theta}{LC} \quad \text{..... (ii)}$$





Here, $C = \frac{m}{100}$

So equation (ii) is

$$S = \frac{10\theta}{L\frac{m}{100}}$$

∴ $$S = \frac{1000\theta}{Lm}$$

Here, m is the mass of sugar in gm

L is the length of tube in cm

θ is the rotational angle in degree

<u>Required Apparatus:</u>

(i) Polarimeter
(ii) Sodium lamp
(iii) Sugar
(iv) Balance
(v) Scale
(vi) Beaker

Observation Table:

Table for angular rotation

Strength of Sugar Solution	No. of Obs.	First reading with water (P)	Second reading with solution (Q)	Angular rotation $\theta = P - Q$	Mean angular rotation (θ)
5%					
10%					
15%					

Calculation:

Vernier Constant = $\frac{1^\circ}{20}$ = 0.05°

Length of the tube, L = 20 cm

For 5% solution, mean angular rotation, θ_1 = …, m_1 = 5 gm

Specific rotation

$$S_1 = \frac{1000\theta_1}{Lm_1}$$

For 10% solution, mean angular rotation, θ_2 = …, m_2 = 10 gm

The specific rotation

$$S_2 = \frac{1000\theta_2}{Lm_2}$$

For angular rotation, θ_3 = …, m_3 = 15 gm

The specific rotation

$$S_3 = \frac{1000\theta_3}{Lm_3}$$

Mean specific rotation, $\theta = \frac{S_1+S_2+S_3}{3}$=….

Result:

We know, the specific rotation of a sugar solution = 60°. Calculated value is ….. °.

Percent Error = $\frac{Standard\ value \sim Calculated\ value}{Standard\ value} x100$ =…… %

Precautions:

1. There should be no air bubble in the tube while filling it with solution or distilled water.
2. While taking one set of the observations, the polarizer should not be disturbed.
3. The cap of the tube should not be tightened beyond a limit as it may strain the glass. Strained glass may produce elliptically polarized light which might interfere with the setting.
4. Two positions at $\pm 90°$ may appear where the equal illumination remains for a long range. These readings should not be taken.
5. Switch off the lamp after completing the experiment.